Encryption

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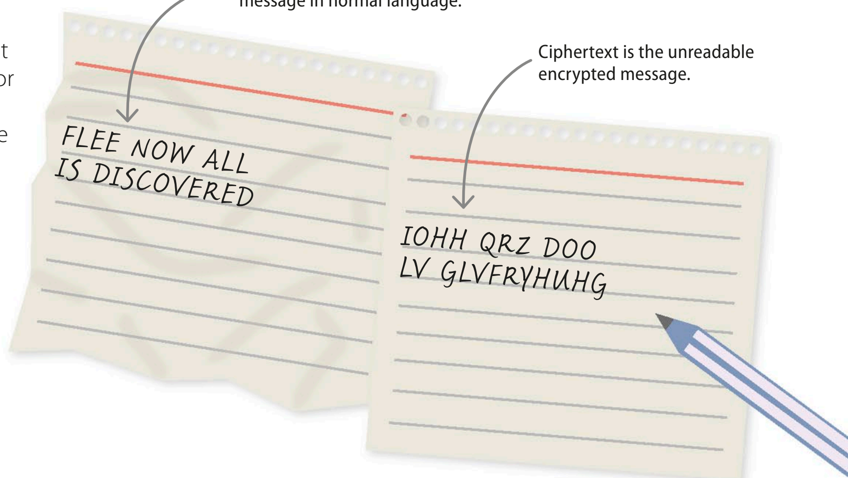
For thousands of years, people have sent messages that could only be understood by the intended recipients, protecting their secrets if the message was intercepted. Creating this kind of message is called encryption.

What is encryption?

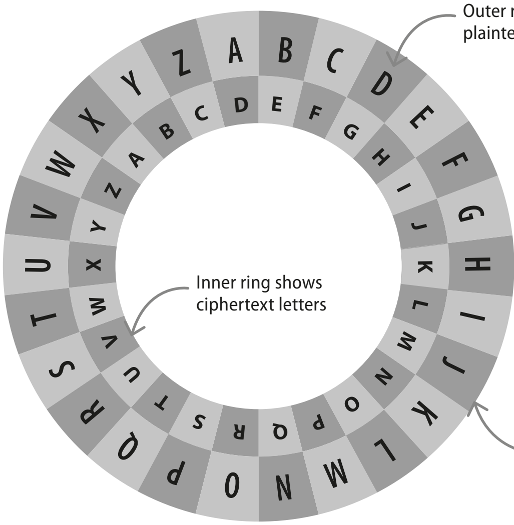
Encryption is the process of taking a message and making it unreadable to everyone except the person it is intended for. Historically, the most popular reason for encrypting information was to allow communication between military leaders, spies, or heads of state. More recently, with the advent of the internet and online shopping, encryption is becoming increasingly important. For instance, it is used to keep shoppers' money safe during transactions.



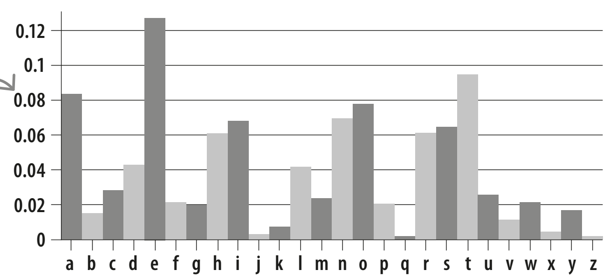
▷ **Plaintext to ciphertext**
Unencrypted information, or plaintext, is encrypted using an algorithm and a key. This generates ciphertext that can be decrypted using the correct key. A cipher is a key to the code.

Early encryption

A transposition cipher changes the position of the letters in a message using a specific rule, called a key. The recipient, who also knows the key, reverses the process to get the original text. Writing a message backwards is a transposition cipher, although it is not a secure one, as it is relatively easy to break the code. Substitution ciphers replace each letter with another letter according to a rule or set of rules.



△ **Substitution cipher**
Used by Julius Caesar, each letter in the substitution cipher is shifted by a set number of spaces along the alphabet. The key is the number of spaces a letter is shifted.



△ **Frequency analysis**
A substitution cipher can be easily broken by frequency analysis. By looking at the encrypted message and finding the most frequent letters, they can be matched to their frequency in the language. The letter "e" is the most frequent in the English language.